Scattering by a Central Potential; the Hamiltonian

HC 6001 – Dr. Castillo

22 October 2024

1 Scattering by a central potential

For a particle of mass m, the incident energy, the incident velocity and the angular momentum are

$$E = \frac{m}{2}\, v_0^2, \tag{1}$$

$$v_0 = \sqrt{2E/m}, \tag{2}$$

$$\ell = s\, m\, v_0 = s\sqrt{2mE}, \tag{3}$$

where s is the impact parameter, so that

$$s = \frac{\ell}{\sqrt{2mE}}. \tag{4}$$

The geometry these quantities describe is set out in Fig. 1.

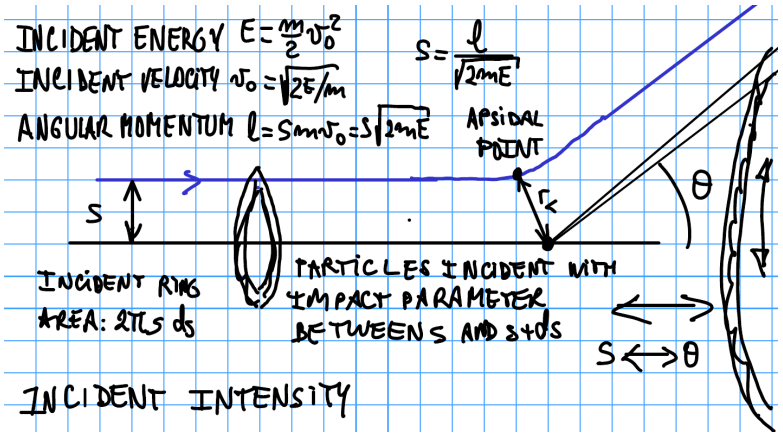


Figure 1: The scattering geometry that fixes the cross section. A particle approaches parallel to the axis at impact parameter s, turns at the apsidal point of closest approach, and leaves along an asymptote making the scattering angle θ with the forward direction; the pair of closely spaced straight lines from the center bracket the interval from θ to $\theta + \mathrm{d}\theta$. Particles entering through the ring of area $2\pi s\,\mathrm{d}s$ are precisely those emerging into that interval, and matching the two counts is what gives Eq. (9). The particle's path is the curve drawn in blue and the axis and bracketing lines are black, a distinction of colour alone, named here in words. The distance of closest approach is marked $r_<$ in the drawing and written r_m from Eq. (18) onward; they are one quantity.

The volume element and the solid angle are

$$\mathrm{d}^3 r = r^2\,\mathrm{d}r\,\underbrace{\mathrm{d}\varphi\,\sin\theta\,\mathrm{d}\theta}_{=\,\mathrm{d}\Omega}, \qquad \mathrm{d}\Omega = 2\pi\sin\theta\,\mathrm{d}\theta, \tag{5}$$

the azimuthal angle φ integrating to 2π because it is irrelevant by symmetry. The quantity to be computed is $s(\theta, E)$.

1.1 Incident intensity and differential cross section

The incident intensity and the differential cross section are defined as ratios of counts:

$$I = \frac{\text{number of particles incident per unit time}}{\text{incident beam cross-sectional area}}, \tag{6}$$

$$\sigma(\Omega)\,\mathrm{d}\Omega = \frac{\text{number of particles scattered into } \mathrm{d}\Omega \text{ per unit time}}{\text{incident intensity}}. \tag{7}$$

For a central potential σ depends on Ω only through θ, so $\sigma(\Omega)$ and $\sigma(\theta)$ denote the same function throughout.

Particles crossing the incident ring between s and $s + \mathrm{d}s$ are exactly those emerging with scattering angle between θ and $\theta + \mathrm{d}\theta$ (Fig. 1), so the two counts are equal:

$$I\,(2\pi s)\,|\mathrm{d}s| = \underset{\text{between } s \text{ and } s+\mathrm{d}s}{\text{particles through the ring}} = \underset{\theta \text{ and } \theta+\mathrm{d}\theta}{\text{particles emerging between}} = I\,\sigma(\theta)\,2\pi\sin\theta\,|\mathrm{d}\theta|. \tag{8}$$

Therefore

$$\boxed{\sigma(\theta) = \frac{s}{\sin\theta}\left|\frac{\mathrm{d}s}{\mathrm{d}\theta}\right|} \tag{9}$$

Define also the total cross section

$$\boxed{\sigma_T = \int \mathrm{d}\Omega\,\sigma(\Omega)} \tag{10}$$

1.2 The deflection angle and the apsidal angle

The two signs of the force give the two geometries of Figs. 2 and 3.

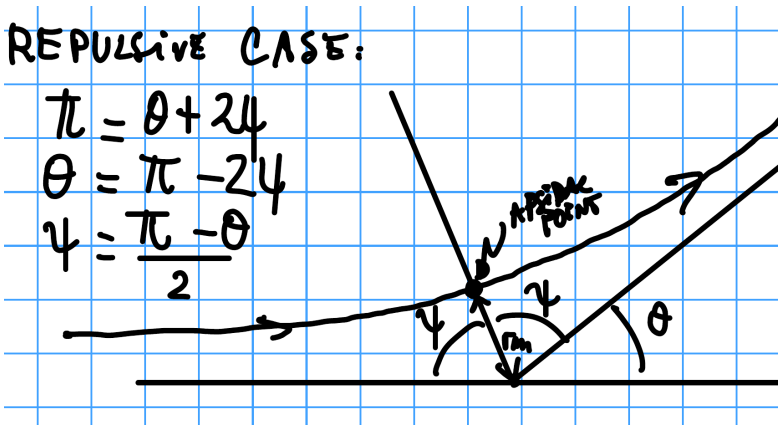


Figure 2: Repulsive deflection. The path turns away from the center at the apsidal point, and the two apsidal angles ψ are measured from the apsidal line, one back to the incoming asymptote and one forward to the outgoing one, which is what makes Eq. (11) a statement about a straight line.

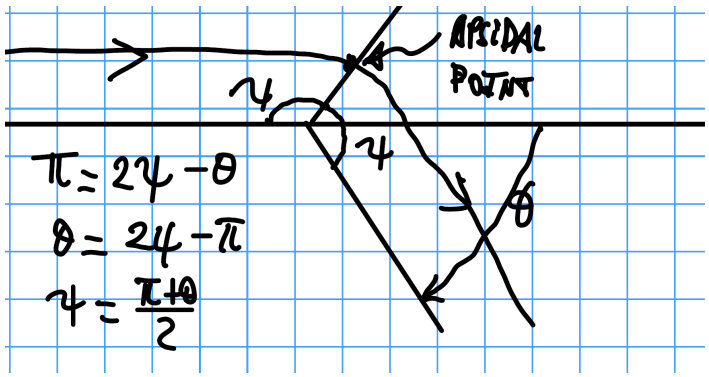


Figure 3: Attractive deflection. The path is pulled across the axis rather than away from it, which reverses how θ is built from the same apsidal angle ψ and accounts for the sign difference between Eqs. (11) and (12).

In the repulsive case

$$\pi = \theta + 2\psi, \qquad \theta = \pi - 2\psi, \qquad \psi = \frac{\pi - \theta}{2}, \tag{11}$$

and in the attractive case

$$\pi = 2\psi - \theta, \qquad \theta = 2\psi - \pi, \qquad \psi = \frac{\pi + \theta}{2}. \tag{12}$$

In all cases

$$\boxed{\theta = |\pi - 2\psi|} \tag{13}$$

1.3 Computing $s(\theta, E)$: the universal method

The radial energy equation and the angular momentum together reduce the orbit to a quadrature. Writing $\tilde{\theta}$ for the running orbit angle, to keep it distinct from the scattering angle θ,

$$E = \frac{m}{2}\dot{r}^2 + V(r) + \frac{\ell^2}{2mr^2}, \tag{14}$$

$$\dot{r} = \left[\frac{2}{m}\left(E - V(r) - \frac{\ell^2}{2mr^2}\right)\right]^{1/2}, \qquad \dot{r} = \frac{\mathrm{d}r}{\mathrm{d}t} \implies \mathrm{d}t = \frac{\mathrm{d}r}{\dot{r}}, \tag{15}$$

and, since $\ell = mr^2\,\frac{\mathrm{d}\tilde{\theta}}{\mathrm{d}t}$,

$$\frac{\mathrm{d}\tilde{\theta}}{\mathrm{d}t} = \frac{\ell}{mr^2} \implies \mathrm{d}\tilde{\theta} = \frac{\ell}{mr^2}\,\mathrm{d}t = \frac{\ell}{mr^2}\,\frac{\mathrm{d}r}{\left[\frac{2}{m}\left(E - V(r) - \frac{\ell^2}{2mr^2}\right)\right]^{1/2}}. \tag{16}$$

Taking $2E/m$ out of the bracket turns this into

$$\mathrm{d}\tilde{\theta} = \underbrace{\frac{\ell}{m}\sqrt{\frac{m}{2E}}}_{= \ell/\sqrt{2mE}\ =\ s}\ \frac{r^{-2}\,\mathrm{d}r}{\left[1 - \frac{V(r)}{E} - \underbrace{\frac{\ell^2}{2mE}}_{=\ s^2}\ \frac{1}{r^2}\right]^{1/2}}, \tag{17}$$

so that, integrating from the apsis out to infinity,

$$\psi = \int_0^{\psi} \mathrm{d}\psi = \int_{\tilde{\theta}_m}^{\pi} \mathrm{d}\tilde{\theta} = \int_{r_m}^{\infty} \frac{s\, r^{-2}\, \mathrm{d}r}{\left[1 - V(r) - \dfrac{s^2}{r^2}\right]^{1/2}}, \tag{18}$$

where $\tilde{\theta}_m$ is the orbit angle at the apsis and r_m the corresponding radius.

Erratum. The potential term in Eq. (18) is written $V(r)$ where $V(r)/E$ is meant. Eq. (17), which it comes from, carries the $/E$, and so does the result it leads to, Eq. (20); without it the bracket adds an energy to two dimensionless terms. Reproduced above as written.

With the substitution

$$u = \frac{1}{r}, \qquad \mathrm{d}u = r^{-2}\, \mathrm{d}r, \qquad u_m = \frac{1}{r_m}, \tag{19}$$

Erratum. $\mathrm{d}u = r^{-2}\, \mathrm{d}r$ is written without its minus sign. The sign is absorbed exactly by the reversal of limits that turns $\int_{r_m}^{\infty}$ into the integral from 0 to u_m below. Reproduced above as written.

Eq. (18) then becomes

$$\boxed{\theta(s) = \left| \pi - 2 \int_0^{u_m} \frac{s\, \mathrm{d}u}{\left[1 - \frac{V(u)}{E} - s^2 u^2\right]^{1/2}} \right|} \tag{20}$$

From here, compute

$$\left|\frac{\mathrm{d}s}{\mathrm{d}\theta}\right| = \frac{1}{\left|\dfrac{\mathrm{d}\theta}{\mathrm{d}s}\right|} \tag{21}$$

and use it to get $\sigma(\theta) = \frac{s}{\sin\theta} \left|\frac{\mathrm{d}s}{\mathrm{d}\theta}\right|$, which is Eq. (9).

Up to here the results are valid for *any* central potential $V(r)$.

2 Coulomb scattering

Charges are written in Gaussian units throughout, as in the reference convention, so that the Coulomb force carries no factor of $4\pi\epsilon_0$. The two like charges of Fig. 4 drive each other apart.

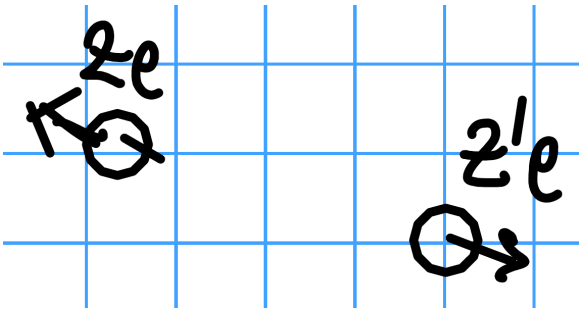


Figure 4: The two charges Ze and $Z'e$ driving each other apart along the line joining them, the case in which the force of Eq. (22) is repulsive.

The force and potential are

$$f = \frac{ZZ'e^2}{r^2}, \qquad f = -\frac{k}{r^2}, \qquad V = -\frac{k}{r}, \qquad k = -ZZ'e^2. \tag{22}$$

For scattering, $E > 0$, and the orbit is

$$\frac{1}{r} = C\left[1 + \varepsilon\cos(\tilde{\theta} - \tilde{\theta}')\right], \qquad C = \frac{mk}{\ell^2} = -\frac{m}{\ell^2}\, ZZ'e^2, \tag{23}$$

with eccentricity

$$\varepsilon = \left[1 + \frac{2E\ell^2}{mk^2}\right]^{1/2} = \left[1 + \frac{2E\ell^2}{m\,(ZZ'e^2)^2}\right]^{1/2} = \left[1 + \left(\frac{2sE}{ZZ'e^2}\right)^2\right]^{1/2} > 1, \tag{24}$$

the last step following from $\ell = s\, m\, v_0 = s\sqrt{2mE}$, which gives

$$\frac{2\ell^2 E}{m} = \frac{s^2\, 2^2\, \cancel{m}\, E^2}{\cancel{m}} = (2sE)^2. \tag{25}$$

The orbit types this potential admits are listed in Table 1.

Table 1: Orbit type by sign of the energy. These three cases hold only for the potential $V = -k/r$.

Energy	Orbit
$E > 0$	hyperbolic
$E = 0$	parabolic
$E < 0$	elliptic

There are two situations:

1. repulsive case, $ZZ' > 0 \implies k = -ZZ'e^2 < 0$;
2. attractive case, $ZZ' < 0 \implies k = -ZZ'e^2 > 0$ (also gravity, $k = Gm_1m_2$).

2.1 Repulsive case

Here $k < 0$ and $C = mk/\ell^2 < 0$, so, dividing by the negative C and reversing the inequality at each step,

$$\begin{aligned} \frac{1}{r} = C\left[1 + \varepsilon\cos(\tilde{\theta} - \tilde{\theta}')\right] > 0 &\iff 1 + \varepsilon\cos(\tilde{\theta} - \tilde{\theta}') < 0 \\ &\iff \varepsilon\cos(\tilde{\theta} - \tilde{\theta}') < -1 \\ &\iff \cos(\tilde{\theta} - \tilde{\theta}') < -\frac{1}{\varepsilon}. \end{aligned} \tag{26}$$

The cosine must be negative and larger in magnitude than $1/\varepsilon$. In this case $\tilde{\theta} = \tilde{\theta}'$ is not a physical point, because $\cos(\tilde{\theta}' - \tilde{\theta}') = \cos 0 = 1$ would give $\frac{1}{r} = C\,[1 + \varepsilon] < 0$, which is unphysical. So $\tilde{\theta} = \tilde{\theta}'$ lies in the middle of the inaccessible region of the hyperbola, and $\tilde{\theta} = \tilde{\theta}' \pm \pi$ is the periapsis; choosing $\tilde{\theta}' = \pi$ gives

$$\cos(\tilde{\theta} - \tilde{\theta}') = \cos(\tilde{\theta} - \pi) = \cos\tilde{\theta}\ \underbrace{\cos\pi}_{-1} + \sin\tilde{\theta}\ \underbrace{\sin\pi}_{0} = -\cos\tilde{\theta}, \tag{27}$$

and therefore

$$\frac{1}{r} = \left(-\frac{mk}{\ell^2}\right)\left[-1-\varepsilon\cos(\tilde{\theta}-\tilde{\theta}')\right] = \frac{m\,ZZ'e^2}{\ell^2}\left[\varepsilon\cos\tilde{\theta}-1\right]. \tag{28}$$

This is a maximum at $\tilde{\theta}=0$, so the minimum radius is

$$\frac{1}{r_m} = \frac{m\,ZZ'e^2}{\ell^2}\left[\varepsilon-1\right], \tag{29}$$

and the angle at the apsidal point is $\tilde{\theta}=0$. From here on the origin of $\tilde{\theta}$ is the apsidal line, so the apsis is at $\tilde{\theta}=0$. In Eq. (18) the origin was the forward beam axis, with the incoming asymptote at $\tilde{\theta}=\pi$.

What is needed now is $\psi = \tilde{\theta}(r\to\infty) = \tilde{\theta}(1/r\to 0)$. In the repulsive case $\theta = \pi - 2\psi$, so $\psi = (\pi-\theta)/2$ and

$$0 = \varepsilon\cos\psi - 1 \iff \frac{1}{\varepsilon} = \cos\psi = \cos\left(\frac{\pi-\theta}{2}\right) = \underbrace{\cos\frac{\pi}{2}}_{0}\cos\frac{\theta}{2} + \underbrace{\sin\frac{\pi}{2}}_{1}\sin\frac{\theta}{2} = \sin\frac{\theta}{2}. \tag{30}$$

2.2 Attractive case

Here $k>0$ and $C = km/\ell^2 > 0$. Take $\tilde{\theta}'=0$; then $\tilde{\theta}=\tilde{\theta}'=0$ gives

$$\frac{1}{r} = C\left[1+\varepsilon\cos(0)\right] = C\left[1+\varepsilon\right], \tag{31}$$

a maximum at $\tilde{\theta}=0$, so the minimum radius is $1/r_m = C\left[1+\varepsilon\right]$ and the apsidal point is also at $\tilde{\theta}=0$.

Again what is needed is $\psi = \tilde{\theta}(r\to\infty) = \tilde{\theta}(1/r\to 0)$. In the attractive case $\theta = 2\psi - \pi$, so $\psi = (\pi+\theta)/2$ and

$$0 = 1+\varepsilon\cos\psi \iff -\frac{1}{\varepsilon} = \cos\psi = \cos\left(\frac{\pi+\theta}{2}\right) = \underbrace{\cos\frac{\pi}{2}}_{0}\cos\frac{\theta}{2} - \underbrace{\sin\frac{\pi}{2}}_{1}\sin\frac{\theta}{2} = -\sin\frac{\theta}{2}. \tag{32}$$

3 The Rutherford cross section

Cancelling the overall minus sign in Eq. (32) gives $1/\varepsilon = \sin(\theta/2)$, exactly as Eq. (30) does, so both cases can be carried forward together.

Supplied. The step joining the two cases is supplied. Each case is carried to its own result and the two are then used together without it.

$$\cot^2\frac{\theta}{2} = \frac{1}{\tan^2\frac{\theta}{2}} = \frac{\cos^2\frac{\theta}{2}}{\sin^2\frac{\theta}{2}} = \frac{1-\sin^2\frac{\theta}{2}}{\sin^2\frac{\theta}{2}} = \frac{1}{\sin^2\frac{\theta}{2}} - 1 = \varepsilon^2 - 1 = \left(\frac{2sE}{ZZ'e^2}\right)^2, \tag{33}$$

so that

$$\boxed{s = s(\theta,E) = \left|\frac{ZZ'e^2}{2E}\right|\cot\frac{\theta}{2}} \tag{34}$$

Differentiating,

$$\frac{\mathrm{d}s}{\mathrm{d}\theta} = \left|\frac{ZZ'e^2}{2E}\right| \frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}}\right) = \left|\frac{ZZ'e^2}{2E}\right| \left(\frac{1}{2}\frac{-\sin\frac{\theta}{2}}{\sin\frac{\theta}{2}} - \frac{1}{2}\frac{\cos^2\frac{\theta}{2}}{\sin^{-2}\frac{\theta}{2}}\right)$$
$$= -\left|\frac{ZZ'e^2}{4E}\right| \frac{1}{\sin^2\frac{\theta}{2}}, \tag{35}$$

Erratum. The second term inside the bracket of Eq. (35) is written over $\sin^{-2}(\theta/2)$ where $\sin^2(\theta/2)$ is meant; the quotient rule gives $-\frac{1}{2}\cot^2(\theta/2)$ for it, which with the first term makes $-\frac{1}{2}\csc^2(\theta/2)$, and that is what the second line uses. Reproduced above as written.

Substituting into Eq. (9) gives

$$\sigma(\theta) = \frac{s}{\sin\theta}\left|\frac{\mathrm{d}s}{\mathrm{d}\theta}\right| = \frac{1}{2}\left(\frac{ZZ'e^2}{2E}\right)^2 \cdot \frac{1}{2\sin\frac{\theta}{2}\cancel{\cos\frac{\theta}{2}}} \cdot \frac{\cancel{\cos\frac{\theta}{2}}}{\sin\frac{\theta}{2}} \cdot \frac{1}{\sin^2\frac{\theta}{2}}, \tag{36}$$

which is the Rutherford scattering cross section for Coulomb forces:

$$\boxed{\sigma(\theta) = \frac{1}{4}\left(\frac{ZZ'e^2}{2E}\right)^2 \csc^4\left(\frac{\theta}{2}\right)} \tag{37}$$

3.1 Remarks on Eq. (37)

1. The cross section is the same for the repulsive and attractive cases, since it is proportional to $(ZZ')^2$.
2. It diverges like E^{-2} at low energies.
3. It diverges like θ^{-4} (very strongly) for small angles θ.
4. This classical result is exactly the same as the result obtained in quantum mechanics.

Erratum. $(zz')^2$ is written in lower case, E^{-2} and θ^{-4} as unrendered markup, and the third remark as "VERY STRONG!!!". The mathematics is set here as mathematics and the casing brought into line with the ZZ' used throughout.

3.2 Total scattering cross section

The total cross section of Eq. (10) may now be evaluated:

$$\sigma_T = \int \sigma(\Omega)\,\mathrm{d}\Omega. \tag{38}$$

In this case $\sigma(\theta) = \text{const} \cdot 1/\sin^4(\theta/2)$, and with

$$\mathrm{d}\Omega = 2\pi\sin\theta\,\mathrm{d}\theta = 4\pi\sin\frac{\theta}{2}\cos\frac{\theta}{2}\,\mathrm{d}\theta, \tag{39}$$

$$\sigma_T = \frac{4\pi}{4}\left(\frac{ZZ'e^2}{2E}\right)^2 \int_0^\pi \frac{\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sin^4\frac{\theta}{2}}\,\mathrm{d}\theta, \tag{40}$$

the range $0 < \theta \leq \pi$ being the full span of deflections available (Fig. 5). The small-angle behaviour of the integrand controls the result:

$$\sigma_T \sim \int_0^\delta \frac{\mathrm{d}\theta}{(\theta/2)^3} = 8 \int_0^\delta \frac{\mathrm{d}\theta}{\theta^3} = 8 \left. \frac{\theta^{-2}}{(-2)} \right|_0^\delta = +\infty, \tag{41}$$

which is strongly divergent.

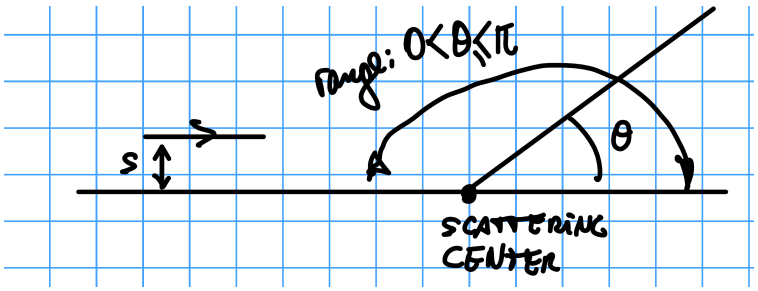


Figure 5: The full range of deflection directions available to a particle passing the scattering center, $0 < \theta \leq \pi$, which is the range integrated over in Eq. (40).

The divergence behaves differently in the two theories, as Table 2 records.

Table 2: Behaviour of the total cross section for long-range potentials in the two theories.

Theory	Total cross section
Classical	$\sigma_T = +\infty$ for any long-range potential
Quantum	$\sigma_T = +\infty$ for some long-range potentials, those whose $V(r)$ does not decay fast enough as $r \to \infty$

3.3 Notation

The source itself compares two of these symbols to Goldstein, Poole & Safko, *Classical Mechanics*, 3rd ed.: $\sigma(\Omega)$ and σ_T, against $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ and σ as written elsewhere. The remaining rows of Table 3 are supplied to extend that same comparison to the other symbols in this document, drawn from Ch. 3.10 and Ch. 8, whose notation might otherwise be read against a different convention.

Table 3: Symbols used here, their counterparts in Goldstein, Poole & Safko, and the departures from that convention.

Used here	Goldstein / elsewhere	Quantity, and any departure
$\sigma(\Omega)$	$\sigma(\Omega)$	differential scattering cross section; matches Goldstein directly ($\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ is common elsewhere)
σ_T	σ_T	total scattering cross section; matches Goldstein directly (σ alone is common elsewhere)
θ	Θ	scattering angle; Goldstein sets it capital, and lower case is kept here
$\tilde{\theta}$	θ	running orbit angle, tilded so that it cannot be read as the scattering angle
ψ	Ψ	angle from the apsidal line to the incoming asymptote (Figs. 2, 3); Goldstein introduces it unnamed and capitalized, lower case is kept here
ε	e	orbit eccentricity, kept distinct from the elementary charge e
s	s	impact parameter; matches Goldstein directly (Taylor's *Classical Mechanics* writes b for the same quantity, noted here only for a reader coming from that text)
q_i, p_i	q_i, p_i	generalized coordinate and conjugate momentum (Ch. 8); match Goldstein directly

Caveat 1. In a dense conducting material the potential is screened,

$$V = -\frac{k}{r} \longrightarrow V^{\text{screen}}(r) = -\frac{k}{r}\,\mathrm{e}^{-r/\lambda}, \tag{42}$$

the screened Coulomb ("Yukawa") potential. In non-conducting materials $V^{\text{screen}}(r)$ may take a different form, but will still vanish much faster than $1/r$.

4 The Hamiltonian and Hamilton's equations of motion

Starting from

$$L = T - V = L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n, t), \tag{43}$$

the Lagrange equations and the conjugate momenta are

$$0 = \frac{\mathrm{d}}{\mathrm{d}t}\left(\left.\frac{\partial L}{\partial \dot{q}_i}\right|_{q_j,\dot{q}_j}\right) - \left.\frac{\partial L}{\partial q_i}\right|_{\dot{q}_j,q_j}, \qquad p_i = \left.\frac{\partial L}{\partial \dot{q}_i}\right|_{q_j,\dot{q}_j}. \tag{44}$$

Remark 1 (Midterm remark). In polar coordinates $\partial L/\partial\dot{\theta} = p_\theta = \ell_z$, where θ is the polar coordinate rather than the scattering angle of the preceding sections. What follows from it is struck out: ~~$L = L(r, \dot{r}, p_\theta)$~~, ~~giving the Lagrange equation~~. The conserved momentum may not be substituted back into the Lagrangian and the Lagrange equations then reused.

4.1 The Legendre transformation

The transformation to be made is

$$q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n \longrightarrow q_1, \dots, q_n, p_1, \dots, p_n, \tag{45}$$

and for a function of two variables it rests on

$$f(x,y) \implies \mathrm{d}f = \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y. \tag{46}$$

4.1.1 The thermodynamic case

In this subsection T, V and p denote temperature, volume and pressure, not the kinetic energy, potential and momenta of the preceding sections. The first law, for a fluid at constant N, is

$$\mathrm{d}U = đQ + đW = T\,\mathrm{d}S - p\,\mathrm{d}V, \qquad U = U(S,V) \text{ the internal energy.} \tag{47}$$

To have T and V as the variables instead, introduce the Helmholtz free energy $F = A = U - TS$:

$$\mathrm{d}A = \mathrm{d}U - \mathrm{d}(TS) = \cancel{T\,\mathrm{d}S} - p\,\mathrm{d}V - \cancel{T\,\mathrm{d}S} - S\,\mathrm{d}T = -S\,\mathrm{d}T - p\,\mathrm{d}V \implies A = A(T,V). \tag{48}$$

4.1.2 The general case

The same transformation in the abstract, with u here the variable conjugate to x rather than the $1/r$ of Eq. (19), reads

$$f(x,y) = u\,\mathrm{d}x + v\,\mathrm{d}y, \qquad u = \frac{\partial f}{\partial x}, \qquad v = \frac{\partial f}{\partial y}, \qquad (x,y) \longrightarrow (u,y). \tag{49}$$

Erratum. The left-hand side of Eq. (49) is written $f(x,y)$ where the differential $\mathrm{d}f$ is meant, as Eq. (46) has it and as Eq. (50) then consumes it. Reproduced above as written.

$$g = f - ux \implies \mathrm{d}g = \mathrm{d}f - \mathrm{d}(ux) = \cancel{u\,\mathrm{d}x} + v\,\mathrm{d}y - \cancel{u\,\mathrm{d}x} - x\,\mathrm{d}u = -x\,\mathrm{d}u + v\,\mathrm{d}y, \qquad g = g(u,y). \tag{50}$$

4.2 The Hamiltonian

Applying this to $L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)$ with

$$\boxed{p_i = \left.\frac{\partial L}{\partial \dot{q}_i}\right|_{q_j, \dot{q}_j}} \tag{51}$$

gives

$$H = -\left(L - \sum_i p_i \dot{q}_i\right), \tag{52}$$

written this way to mirror $g = f - ux$ of Eq. (50), with $H = -g$; it is the usual $\sum_i p_i \dot{q}_i - L$. Then

$$\mathrm{d}H = -\mathrm{d}L + \sum_i \mathrm{d}(p_i \dot{q}_i) = -\frac{\partial L}{\partial t} - \sum_i \left(\frac{\partial L}{\partial q_i}\,\mathrm{d}q_i + \cancel{\frac{\partial L}{\partial \dot{q}_i}\,\mathrm{d}\dot{q}_i}\right) + \sum_i \left(\cancel{p_i\,\mathrm{d}\dot{q}_i} + \dot{q}_i\,\mathrm{d}p_i\right), \tag{53}$$

so that

$$\mathrm{d}H = -\frac{\partial L}{\partial t} + \sum_i \left(\dot{q}_i\,\mathrm{d}p_i - \frac{\partial L}{\partial q_i}\,\mathrm{d}q_i\right) \implies H = H(p_1, \dots, p_n, q_1, \dots, q_n, t). \tag{54}$$

Erratum. The first term of Eqs. (53) and (54) carries no differential: with every other term a differential, it is $-\frac{\partial L}{\partial t}\,\mathrm{d}t$. Reproduced above as written.

Reading off the coefficients gives the canonical equations of motion, the middle one using the Lagrange equation (44):

$$\boxed{\frac{\partial H}{\partial p_i} = \dot{q}_i, \qquad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} = -\dot{p}_i, \qquad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}} \tag{55}$$